

CAM – MIDTERM EXAM 06/11/2020

EXAMPLES OF SHORT ESSAY (MAX 400 WORDS)

Referring to metal casting, discuss why and how to mitigate the shrinkage.

Example 1 (399 words)

"Shrinkage is a phenomenon that occurs in metal casting processes during the cooling at liquid state, the solidification and at solid state, due to the fact that, when the temperature of the metal is reduced, its volume is reduced, and its density is increased. Therefore, when designing the pattern and the core in order to obtain the cavity, this effect has to be taken into account in order to have an object of the desired dimensions when the casting is extracted from the mold. First of all, shrinkage at solid state causes the reduction of the volume of the object, so it is necessary to increase all the dimensions of the pattern and of the core with respect to the ones of the final object. Given s , which is the percentage of shrinkage, any dimension of the pattern will be computed as: $D_{p_1} = D_p \cdot (1+s)$. Moreover, shrinkage also occurs when the material is cooling at liquid state and during solidification. In those moments, the formation of shrinkage cavities may occur: in fact, the first part of the liquid to solidify will be the one in contact with the mold, causing a lowering of the level of the liquid in the cavity; this process will occur many times, as solidification proceeds towards the inner part of the casting, causing the formation of a cavity (defect in the final product). In order to avoid the formation of such a defect, some other cavities which are not part of the final object will be designed, which will be in contact with the original cavity: they are the so called "risers". They are a reservoir of liquid metal which is supposed to fill the eventual shrinkage cavities in the final object while it solidifies, in order to transfer the defect from the casting to the riser itself. The shape and position of the risers have to be chosen carefully: in fact, they have to be the last elements to solidify in the mold and they should be placed in points in which eventual shrinkage cavities cannot be filled by the liquid metal which is inside other parts of the final object. This means studying the directional solidification of the casting: solidification should proceed from the parts with low volume-to-area ratio to the risers, which have to be such that their ratios are the highest ones, because a higher ratio means longer solidification time."

Example 2 (399 words)

"In metal casting, there are three phases in which shrinkage occurs: 1) Cooling of the liquid state 2) Solidification 3) Cooling of the solid state in each of them, the volume of the casting is reducing since the solid phase has a higher density with respect to the liquid one. In the third one, the shrinkage can be approximated to a linear shrinkage and, as a matter of fact, to mitigate it, it's sufficient to enlarge the dimension of the mold cavity. The first and the second one will cause the formation of a shrinkage cone. To avoid this problem, risers are needed. A riser is a reservoir of molten metal that can supply the region of the mold subjected to shrinkage. It's important to design the risers such that the solidification of the metal in them occur after the solidification of the metal in the cavity. This goal can be achieved by setting properly the V/A ratio because the Total Solidification Time can be computed, according to Chvorinov's rule, as: $T = C_m \cdot (V/A)^n$ where C_m depends on the mold's material and n is equal to 2 for many metals. As a consequence, it's important to study the directional solidification of the object that means what are the regions of the casting that solidify first and what ones solidify later. Knowing that, I have to put risers upon or next to the regions that solidify as the last ones since in them I will have the shrinkage cone (because they supply molten metal to the others). If it's not possible to put a riser in that region, it's possible to put some chills (internal or external) that are a way to fasten the heat exchange in particular directions (and so fasten the solidification). In addition to this, I can also put a metal object inside these regions. When the molten metal reaches it, it has to be melted and so energy is consumed, and solidification is quicker. Furthermore, I can try to modify the geometry of these regions in order to lower down $M=V/A$ ("modulo termico"), by increasing the Area in which heat exchange happens or /and by decreasing the volume (design for casting). As a last consideration, it's important remarking that the supply of molten metal from region 1 to region 2 can happen only if $M1/M2$ is between 1,1 and 1,3 (stresses are created)."